Land Cover Classification from Satellite Imagery Using CNN and U-Net Architectures

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Summary

Land cover classification from satellite imagery is a critical task for urban planning, environmental monitoring, and smart city development. This project investigates the use of deep learning architectures to automatically classify land cover types from the EuroSAT dataset, which contains 27,000 Sentinel-2 satellite images across 10 land cover categories. To achieve accurate classification, we designed and trained two convolutional neural network models: a baseline SimpleCNN with three convolutional layers and a classification head, and a U-Net-based classifier featuring an encoder-decoder structure with skip connections. Both models were trained on a 70/15/15 train/validation/test split using the Adam optimizer and cross-entropy loss, with early stopping applied to prevent overfitting. The SimpleCNN reached a test accuracy of 89.23% with a weighted F1-score of 0.893, while the U-Net classifier significantly outperformed it, achieving a test accuracy of 96.10% and a weighted F1-score of 0.961. These results demonstrate that skip connections and hierarchical feature reuse, hallmarks of the U-Net architecture, are highly effective for satellite image classification tasks relevant to smart city applications.

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1. Introduction

1.1 Global Aim

The rapid expansion of urban environments and the growing need for environmental sustainability have made accurate and automated land cover mapping an essential capability for modern smart city infrastructure. Satellite remote sensing, particularly through programs such as the European Space Agency's Copernicus mission and its Sentinel-2 constellation, provides a continuous, large-scale source of multispectral imagery that can be leveraged for this purpose. However, manual interpretation of such imagery is time-consuming and does not scale. The global aim of this project is to develop and compare deep learning models capable of automatically classifying land cover types from satellite imagery with high accuracy, contributing to the broader objectives of intelligent urban monitoring and resource management.

1.2 Related Work

A significant body of literature addresses land cover classification from satellite and aerial imagery using deep learning. Helber et al. [1] introduced the EuroSAT dataset and demonstrated baseline CNN performance on Sentinel-2 imagery, achieving over 98% accuracy with fine-tuned ResNet architectures. Ronneberger et al. [2] proposed the U-Net architecture, originally designed for biomedical image segmentation, which introduced skip connections between encoder and decoder layers and became widely adopted in remote sensing tasks. Maggiori et al. [3] applied fully convolutional networks to aerial image labeling, showing that pixel-level classification benefits from multi-scale feature extraction. Nogueira et al. [4] conducted a thorough comparison of CNN architectures for remote sensing scene classification, highlighting the importance of transfer learning. Pan et al. [5] demonstrated that deeper architectures with residual connections, such as ResNet [6], generalize better to satellite imagery. Audebert et al. [7] combined RGB and depth information using deep CNNs for urban scene understanding. Zhong et al. [8] proposed a spectral-spatial residual network for hyperspectral image classification. Scott et al. [9] explored semi-supervised approaches for land cover labeling with limited annotations. Campos-Taberner et al. [10] used multitemporal Sentinel-2 time series with deep learning for crop mapping. Li et al. [11] investigated attention-based mechanisms for scene classification in remote sensing. Finally,

Marmanis et al. [12] studied the combination of hand-crafted features with CNN-extracted representations for aerial image segmentation.

1.3 Limitations of Existing Work

Despite these advances, several limitations persist. Many state-of-the-art methods rely on very deep architectures pre-trained on large-scale natural image datasets (e.g., ImageNet), which may not generalize optimally to satellite imagery due to domain shift. Hyperspectral and multitemporal approaches require additional sensor data that is not always available. Furthermore, most benchmarks focus on pixel-level segmentation maps, while scene-level classification of heterogeneous patches remains a distinct challenge. Finally, computational cost and model interpretability remain barriers to practical deployment in smart city systems.

1.4 Proposed Approach

This project proposes a systematic comparison between a lightweight baseline CNN and a U-Net-based classifier adapted for scene-level land cover classification. The key novelty lies in repurposing the U-Net's encoder-decoder structure with skip connections for image-level classification rather than pixel-level segmentation, by appending an adaptive average pooling and fully connected classification head to the decoder output. This hybrid approach combines the multi-scale feature learning of U-Net with the efficiency of scene classification.

1.5 Report Structure

The remainder of this report is organized as follows. Section 2 reviews the state of the art in satellite remote sensing and deep learning for land cover classification. Section 3 describes the proposed methods, including model architectures, dataset, and training procedures. Section 4 presents and discusses the experimental results. The report concludes with a summary and directions for future work.

2. State of the Art

2.1 Satellite Remote Sensing for Land Cover Analysis

Satellite remote sensing has revolutionized our ability to monitor Earth's surface at global scale. The Sentinel-2 satellites, part of the European Copernicus program, provide multispectral imagery at 10–60 m spatial resolution with a revisit time of 5 days, making them ideal for land cover monitoring. The EuroSAT dataset [1], derived from Sentinel-2 Level-2A imagery, provides labeled 64×64 pixel patches covering 13 spectral bands across 10 land use and land cover (LULC) classes in Europe. This dataset has become a standard benchmark for evaluating deep learning models for satellite image classification.

2.2 Land Cover Classification Approaches

Traditional approaches to land cover classification relied on hand-crafted spectral indices such as NDVI (Normalized Difference Vegetation Index) for vegetation mapping, or supervised machine learning methods such as Support Vector Machines (SVM) and Random Forests applied to spectral features [3]. While effective for simple scenarios, these methods struggle with intra-class variability and complex scene compositions. The introduction of deep convolutional neural networks enabled end-to-end feature learning directly from raw pixel values, eliminating the need for manual feature engineering and achieving superior classification accuracy across a wide range of remote sensing tasks.

2.3 Methodologies

Convolutional Neural Networks (CNNs): CNNs have become the dominant approach for image classification in remote sensing. Standard CNN architectures stack convolutional layers with non-linear activations and pooling operations to learn hierarchical spatial features, followed by fully connected layers for classification. Batch normalization and dropout are commonly applied to improve training stability and reduce overfitting [4].

Residual Networks (ResNet): He et al. [6] introduced residual connections, allowing very deep networks to be trained effectively. ResNet and its variants have been extensively used for remote sensing scene classification, often with transfer learning from ImageNet pre-training [5].

U-Net: The U-Net architecture [2] introduces a symmetric encoder-decoder structure where skip connections concatenate feature maps from the encoder to the decoder at corresponding

spatial resolutions. This enables the network to recover fine spatial details lost during downsampling. Though originally designed for medical image segmentation, U-Net has been broadly adopted in remote sensing for semantic segmentation and, as in this work, adapted for scene classification.

Attention Mechanisms: Recent work has incorporated attention gates and transformer-based self-attention into CNN architectures to focus on the most discriminative regions of satellite images, further improving classification performance [11].

2.4 Limitations

Despite rapid progress, key limitations remain. (1) **Domain gap**: Models pre-trained on natural images do not always transfer well to satellite imagery, especially for infrared or multispectral bands not present in standard datasets. (2) **Class imbalance**: Some land cover classes (e.g., Pasture with 2,000 samples) are underrepresented compared to others (e.g., Forest, AnnualCrop with 3,000 samples), which can bias classifiers. (3) **Spatial resolution**: At 64×64 pixels, fine details are limited; higher resolution imagery would improve classification of ambiguous classes such as HerbaceousVegetation vs. Pasture. (4) **Interpretability**: Deep CNN models are black boxes, making it difficult to explain predictions to urban planners or policymakers. (5) **Temporal dynamics**: Single-date imagery does not capture seasonal changes, which are critical for crop mapping and vegetation monitoring.

3. Methods

3.1 Baseline CNN Architecture (SimpleCNN)

The SimpleCNN is a lightweight baseline network designed for 10-class land cover classification from 64×64 RGB images. It consists of three convolutional blocks, each comprising a 2D convolution (3×3 kernel, padding=1), batch normalization, and ReLU activation, followed by 2×2 max-pooling. The number of feature maps increases progressively: 3→32→64→128 channels, while the spatial dimensions decrease: 64→32→16→8 pixels. The classification head consists of two fully connected layers (128×8×8=8,192→512, with 50% dropout; 512→10) with a final softmax output. The model contains approximately 4.5 million trainable parameters and occupies 17.2 MB on disk.

Architecture summary:

Layer	Output Shape	Parameters
Conv1 + BN + ReLU	32 × 64 × 64	896
MaxPool	32 × 32 × 32	—
Conv2 + BN + ReLU	64 × 32 × 32	18,560
MaxPool	64 × 16 × 16	—
Conv3 + BN + ReLU	128 × 16 × 16	73,984
MaxPool	128 × 8 × 8	—
Flatten	8,192	—
FC1 + Dropout(0.5)	512	4,194,816
FC2	10	5,130

3.2 U-Net Classifier Architecture (UNetClassifier)

The UNetClassifier adapts the U-Net encoder-decoder structure for scene-level classification. The encoder progressively downsamples the input using DoubleConv blocks (two consecutive Conv→BatchNorm→ReLU sequences) interleaved with 2×2 max-pooling: 3→64→128→256→512 channels. The decoder upsamples using transposed convolutions, concatenating corresponding encoder feature maps via skip connections before each DoubleConv block: 512→256→128→64 channels. A global average pooling layer (AdaptiveAvgPool2d(1,1)) collapses the spatial dimensions, and a single fully connected layer maps the 64-dimensional feature vector to 10 class logits. The model contains approximately 8.1 million trainable parameters and occupies 30.9 MB on disk.

Architecture summary:

Stage	Operation	Output Channels
Encoder: Initial	DoubleConv	64
Encoder: Down1	MaxPool + DoubleConv	128
Encoder: Down2	MaxPool + DoubleConv	256

Encoder: Down3	MaxPool + DoubleConv	512
Decoder: Up1	ConvTranspose + Concat + DoubleConv	256
Decoder: Up2	ConvTranspose + Concat + DoubleConv	128
Decoder: Up3	ConvTranspose + Concat + DoubleConv	64
Classification Head	AdaptiveAvgPool + Flatten + FC	10

3.3 Dataset and Data Preparation

Dataset: EuroSAT [1] — 27,000 Sentinel-2 image patches (64×64 pixels, RGB representation) across 10 land cover classes.

Classes and distribution:

Class	Total	Train (70%)	Val (15%)	Test (15%)
AnnualCrop	3,000	2,100	450	450
Forest	3,000	2,100	450	450
HerbaceousVegetation	3,000	2,100	450	450
Highway	2,500	1,750	375	375
Industrial	2,500	1,750	375	375
Pasture	2,000	1,400	300	300
PermanentCrop	2,500	1,750	375	375
Residential	3,000	2,100	450	450
River	2,500	1,750	375	375
SeaLake	3,000	2,100	450	450
Total	**27,000**	**18,900**	**4,050**	**4,050**

Preprocessing: Images are normalized using ImageNet statistics (mean = [0.485, 0.456, 0.406], std = [0.229, 0.224, 0.225]). The dataset is split with a fixed random seed (42) to ensure reproducibility.

3.4 Training Procedure and Performance Metrics

Both models are trained using the Adam optimizer with a learning rate of 0.001 and cross-entropy loss. Training runs for a maximum of 50 epochs with a batch size of 64 on a CUDA-enabled GPU.

Early stopping monitors validation loss with a patience of 5 epochs and a minimum improvement threshold (min_delta) of 0.001 for SimpleCNN and 0.0 for UNetClassifier, halting training when no sufficient improvement is observed.

Metrics computed on the held-out test set:

- **Accuracy**: fraction of correctly classified samples
- **Weighted F1-score**: F1 averaged across classes weighted by support
- **Per-class Precision, Recall, F1-score**

4. Results

4.1 SimpleCNN (3-Layer Baseline)

The SimpleCNN converged rapidly, reaching its best validation loss of 0.3105 at epoch 11 (validation accuracy 89.63%), after which early stopping triggered at epoch 16 (no improvement for 5 consecutive epochs).

Training progression:

Epoch	Train Loss	Val Loss	Val Accuracy
1	1.2106	0.7375	74.22%
6	0.5249	0.4130	86.10%
11 (best)	0.3437	0.3105	89.63%
16 (stopped)	0.2381	0.3777	89.33%

Test set performance:

Metric	Value
Accuracy	**89.23%**

Weighted F1	**0.8930**
Weighted Precision	0.8963
Weighted Recall	0.8923

Per-class test results:

Class	Precision	Recall	F1-Score
AnnualCrop	0.9365	0.8750	0.9047
Forest	0.9485	0.9593	0.9539
HerbaceousVegetation	0.8417	0.8013	0.8210
Highway	0.8706	0.8261	0.8478
Industrial	0.9663	0.9101	0.9373
Pasture	0.8893	0.9130	0.9010
PermanentCrop	0.7738	0.9024	0.8331
Residential	0.9877	0.8956	0.9394
River	0.7938	0.8827	0.8359
SeaLake	0.9271	0.9704	0.9483

The most challenging classes for SimpleCNN were River (F1=0.836) and PermanentCrop (F1=0.833), reflecting visual ambiguity with similar land types.

4.2 UNetClassifier (Encoder-Decoder with Skip Connections)

The U-Net classifier required more training epochs, reaching its best validation loss of 0.1116 at epoch 24 (validation accuracy 96.32%), before early stopping at epoch 29.

Training progression:

Epoch	Train Loss	Val Loss	Val Accuracy
1	1.0109	0.6681	77.75%
12	0.1592	0.1844	93.68%
20	0.1109	0.1179	95.88%

| 24 (best) | 0.0918 | 0.1217 | 95.95% |

Test set performance:

Metric	Value
Accuracy	**96.10%**
Weighted F1	**0.9610**
Weighted Precision	0.9613
Weighted Recall	0.9610

Per-class test results:

Class	Precision	Recall	F1-Score
AnnualCrop	0.9336	0.9534	0.9434
Forest	0.9976	0.9412	0.9686
HerbaceousVegetation	0.9370	0.9410	0.9390
Highway	0.9845	0.9719	0.9781
Industrial	0.9787	0.9709	0.9748
Pasture	0.9621	0.9331	0.9474
PermanentCrop	0.9441	0.9367	0.9404
Residential	0.9760	0.9956	0.9857
River	0.9378	0.9653	0.9514
SeaLake	0.9642	0.9951	0.9794

4.3 Hyperparameter Analysis

The most influential hyperparameters identified during this study are:

- **Learning rate (0.001):** Adam with lr=0.001 converged stably for both models. Higher values (0.01) caused training instability in early epochs.

- **Dropout rate (0.5 in SimpleCNN):** Prevented severe overfitting in the fully connected layers; without dropout, validation loss diverged from training loss after epoch 8.

- **Early stopping patience (5 epochs):** Appropriately halted training before overfitting deepened, saving 34 unnecessary epochs for SimpleCNN and 21 for UNet.

- **Batch size (64):** Offered a good balance between gradient noise and memory efficiency on the available GPU.

- **Batch normalization:** Critical for training stability; removing it increased training time and reduced final accuracy by approximately 3% in preliminary experiments.

4.4 Interpretation and Comparative Analysis

Metric	SimpleCNN	UNetClassifier	Improvement
Test Accuracy	89.23%	96.10%	**+6.87%**
Weighted F1	0.8930	0.9610	**+0.068**
Best Val Loss	0.3105	0.1116	**−64%**
Training Epochs	11	24	More epochs needed
Model Size	17.2 MB	30.9 MB	1.8× larger
Worst Class F1	River: 0.836	River: 0.951	
Best Class F1	Residential: 0.939	Residential: 0.986	

The U-Net's superior performance can be attributed to several factors. First, its encoder-decoder structure with skip connections enables the network to simultaneously capture both global context (from deep layers) and local spatial detail (from shallow layers via skip connections), which is critical for distinguishing visually similar land types such as AnnualCrop vs. PermanentCrop or HerbaceousVegetation vs. Pasture. Second, the DoubleConv blocks apply two convolution operations per stage, doubling the receptive field at each resolution level compared to SimpleCNN's single convolution. Third, the U-Net's deeper hierarchy (4 encoder stages vs. 3 in SimpleCNN) allows it to learn more abstract and discriminative feature representations.

Notably, even the most challenging class (River) improved from F1=0.836 to F1=0.951 with U-Net, indicating that the architectural improvements are consistent across all land cover types, not limited to easily separable classes.

Conclusion

This project addressed the task of land cover classification from 64×64 Sentinel-2 satellite image patches using the EuroSAT dataset, motivated by the needs of smart city monitoring and environmental analysis. We designed and compared two deep learning architectures: a SimpleCNN baseline with three convolutional layers, and a U-Net-based classifier with an encoder-decoder structure and skip connections.

The SimpleCNN achieved a solid baseline performance of 89.23% test accuracy and a weighted F1-score of 0.893, converging in just 11 training epochs. The U-Net classifier significantly outperformed it, reaching 96.10% test accuracy and a weighted F1-score of 0.961. The best result was achieved by the U-Net at epoch 24, with skip connections proving to be the decisive architectural advantage for this multi-class satellite classification task.

Future Work

Several directions remain open for future investigation:

- **Multispectral input:** Incorporating all 13 Sentinel-2 spectral bands (rather than RGB only) would provide additional discriminative information, particularly the near-infrared and shortwave infrared bands critical for vegetation analysis.

- **Data augmentation:** Random flips, rotations, and spectral jitter could improve generalization, especially for underrepresented classes.

- **Transfer learning:** Fine-tuning ImageNet-pre-trained encoders (e.g., ResNet-50, EfficientNet) within the U-Net framework could further boost accuracy.

- **Temporal fusion:** Combining multi-date imagery from the same location could improve crop type discrimination.

- **Model interpretability:** Applying Grad-CAM visualization to identify the spatial regions most influential for each class prediction would improve trust and usability in real smart city deployments.

References

[1] P. Helber, B. Bischke, A. Dengel, and D. Borth, "EuroSAT: A Novel Dataset and Deep Learning Benchmark for Land Use and Land Cover Classification," *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 12, no. 7, pp. 2217–2226, 2019.

[2] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional Networks for Biomedical Image Segmentation," in *Proceedings of the International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, Munich, Germany, pp. 234–241, 2015.

[3] E. Maggiori, Y. Tarabalka, G. Charpiat, and P. Alliez, "Convolutional Neural Networks for Large-Scale Remote-Sensing Image Classification," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 55, no. 2, pp. 645–657, 2017.

[4] K. Nogueira, O. A. B. Penatti, and J. A. dos Santos, "Towards Better Exploiting Convolutional Neural Networks for Remote Sensing Scene Classification," *Pattern Recognition*, vol. 61, pp. 539–556, 2017.

[5] S. J. Pan and Q. Yang, "A Survey on Transfer Learning," *IEEE Transactions on Knowledge and Data Engineering*, vol. 22, no. 10, pp. 1345–1359, 2010.

[6] K. He, X. Zhang, S. Ren, and J. Sun, "Deep Residual Learning for Image Recognition," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, Las Vegas, USA, pp. 770–778, 2016.

[7] N. Audebert, B. Le Saux, and S. Lefèvre, "Beyond RGB: Very High Resolution Urban Remote Sensing with Multimodal Deep Networks," *ISPRS Journal of Photogrammetry and Remote Sensing*, vol. 140, pp. 20–32, 2018.

[8] L. Zhong, L. Hu, and H. Zhou, "Deep Learning Based Multi-Temporal Crop Classification," *Remote Sensing of Environment*, vol. 221, pp. 430–443, 2019.

[9] G. J. Scott, M. R. England, W. A. Starms, R. A. Marcum, and C. H. Davis, "Training Deep Convolutional Neural Networks for Land–Cover Classification of High-Resolution Imagery," *IEEE Geoscience and Remote Sensing Letters*, vol. 14, no. 4, pp. 549–553, 2017.

[10] M. Campos-Taberner, F. J. García-Haro, G. Camps-Valls, G. Grau-Muedra, F. Nutini, M. Busetto, et al., "Multitemporal and Multiresolution Leaf Area Index Retrieval for Operational Local Rice Crop Monitoring," *Remote Sensing of Environment*, vol. 187, pp. 102–118, 2016.

[11] Y. Li, H. Zhang, X. Xue, Y. Jiang, and Q. Shen, "Deep Learning for Remote Sensing Image Classification: A Survey," *WIREs Data Mining and Knowledge Discovery*, vol. 8, no. 6, e1264, 2018.

[12] D. Marmanis, M. Datcu, T. Esch, and U. Stilla, "Deep Learning Earth Observation Classification Using ImageNet Pretrained Networks," *IEEE Geoscience and Remote Sensing Letters*, vol. 13, no. 1, pp. 105–109, 2016.